

When you do:

```
from sklearn import metrics
```

`metrics` is a module containing many functions and submodules for evaluating models. Below is a **structured list of the main methods and objects available in `sklearn.metrics`**, grouped by task.

◆ Classification Metrics

Accuracy & Error

- `accuracy_score`
- `balanced_accuracy_score`
- `zero_one_loss`

Precision / Recall / F1

- `precision_score`
- `recall_score`
- `f1_score`
- `precision_recall_fscore_support`

Confusion & Reports

- `confusion_matrix`
- `ConfusionMatrixDisplay`
- `classification_report`

ROC / AUC

- `roc_curve`
- `roc_auc_score`
- `auc`
- `RocCurveDisplay`
- `DetCurveDisplay`
- `det_curve`

Log & Probability-Based

- `log_loss`
- `brier_score_loss`
- `hinge_loss`

Ranking & Multilabel

- `label_ranking_average_precision_score`
- `label_ranking_loss`
- `coverage_error`
- `average_precision_score`
- `top_k_accuracy_score`

Multiclass / Multilabel

- `jaccard_score`
 - `hamming_loss`
 - `cohen_kappa_score`
 - `matthews_corrcoef`
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◆ Regression Metrics

- `mean_squared_error`
 - `mean_absolute_error`
 - `mean_absolute_percentage_error`
 - `mean_squared_log_error`
 - `median_absolute_error`
 - `max_error`
 - `r2_score`
 - `explained_variance_score`
 - `root_mean_squared_error` (*newer versions*)
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◆ Clustering Metrics

Label-based

- `adjusted_rand_score`
- `rand_score`
- `adjusted_mutual_info_score`
- `normalized_mutual_info_score`
- `mutual_info_score`
- `v_measure_score`
- `fowlkes_mallows_score`
- `homogeneity_score`
- `completeness_score`

Distance-based

- `silhouette_score`
 - `silhouette_samples`
 - `calinski_harabasz_score`
 - `davies_bouldin_score`
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◆ Pairwise Metrics & Distances

Similarity / Distance

- `pairwise_distances`
 - `pairwise_distances_argmin`
 - `pairwise_distances_argmin_min`
 - `pairwise_kernels`
 - `cosine_similarity`
 - `cosine_distances`
 - `euclidean_distances`
 - `manhattan_distances`
 - `nan_euclidean_distances`
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◆ Model Evaluation Utilities

- `get_scorer`
 - `get_scorer_names`
 - `make_scorer`
 - `check_scoring`
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◆ Display Objects (Visualization Helpers)

- `PrecisionRecallDisplay`
 - `RocCurveDisplay`
 - `DetCurveDisplay`
 - `ConfusionMatrixDisplay`
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◆ Submodules

- `metrics.cluster`

- `metrics.pairwise`
 - `metrics.scorer`
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◆ How to see everything programmatically

```
import sklearn.metrics as m
print([name for name in dir(m) if not name.startswith("_")])
```

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